

3 Geometry

What students need to learn:

Ref	Content descriptor	Concepts and skills
GM a	Recall and use properties of angles at a point, angles on a straight line (including right angles), perpendicular lines, and opposite angles at a vertex	- Recall and use properties of angles - angles at a point - angles at a point on a straight line, including right angles - perpendicular lines - vertically opposite angles
GM b	Understand and use the angle properties of parallel lines, triangles and quadrilaterals	- Distinguish between scalene, isosceles, equilateral, and right-angled triangles - Understand and use the angle properties of triangles - Use the angle sum of a triangle is 180° - Understand and use the angle properties of intersecting lines - Understand and use the angle properties of parallel lines - Mark parallel lines on a diagram - Use the properties of corresponding and alternate angles - Understand and use the angle properties of quadrilaterals - Give reasons for angle calculations - Explain why the angle sum of a quadrilateral is 360° - Understand the proof that the angle sum of a triangle is 180° - Understand a proof that the exterior angle of a triangle is equal to the sum of the interior angles at the other two vertices - Use the size/angle properties of isosceles and equilateral triangles Recall and use these properties of angles in more complex problems

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GM c	Calculate and use the sums of the interior and exterior angles of polygons	- Calculate and use the sums of the interior angles of polygons - Use geometric language appropriately and recognise and name pentagons, hexagons, heptagons, octagons and decagons - Use the angle sums of irregular polygons - Calculate and use the angles of regular polygons - Use the sum of the interior angles of an n-sided polygon - Use the sum of the exterior angles of any polygon is 360° - Use the sum of the interior angle and the exterior angle is 180° - Find the size of each interior angle or the size of each exterior angle or the number of sides of a regular polygon - Understand tessellations of regular and irregular polygons - Tessellate combinations of polygons - Explain why some shapes tessellate when other shapes do not
GM d	Recall the properties and definitions of special types of quadrilateral, including square, rectangle, parallelogram, trapezium, kite and rhombus	- Recall the properties and definitions of special types of quadrilateral, including symmetry properties - List the properties of each, or identify (name) a given shape - Classify quadrilaterals by their geometric properties
GM e	Recognise reflection and rotation symmetry of 2-D shapes	- Recognise reflection symmetry of 2-D shapes - Identify and draw lines of symmetry on a shape - Recognise rotation symmetry of 2-D shapes - Identify the order of rotational symmetry of a 2-D shape - Draw or complete diagrams with a given number of lines of symmetry - State the line symmetry as a simple algebraic equation - Draw or complete diagrams with a given order of rotational symmetry

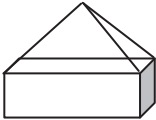
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GM f	Understand congruence and similarity	- Recognise that all corresponding angles in similar figures are equal in size when the lengths of sides are not Understand and use SSS, SAS, ASA and RHS conditions to prove the congruence of triangles using formal arguments, and to verify standard ruler and a pair of compasses constructions Understand similarity of triangles and of other plane figures, and use this to make geometric inferences Complete a formal geometric proof of similarity of two given triangles
GM g	Use Pythagoras' theorem in 2-D and 3-D	- Understand, recall and use Pythagoras' theorem in 2-D, then in 3-D problems Understand the language of planes, and recognise the diagonals of a cuboid Calculate the length of a diagonal of a cuboid
GM h	Use the trigonometric ratios and the sine and cosine rules to solve 2-D and 3-D problems	Use the trigonometric ratios to solve 2-D and 3-D problems Understand, recall and use trigonometric relationships in right-angled triangles, and use these to solve problems in 2-D and in 3-D configurations Find the angle between a line and a plane (but not the angle between two planes or between two skew lines) Find angles of elevation and angles of depression Use the sine and cosine rules to solve 2-D and 3-D problems
GM i	Distinguish between centre, radius, chord, diameter, circumference, tangent, arc, sector and segment	- Recall the definition of a circle and identify (name) and draw the parts of a circle - Understand related terms of a circle - Draw a circle given the radius or diameter

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GM j	Understand and construct geometrical proofs using circle theorems	• Understand and use the fact that the tangent at any point on a circle is perpendicular to the radius at that point • Understand and use the fact that tangents from an external point are equal in length • Find missing angles on diagrams • Give reasons for angle calculations involving the use of tangent theorems • Prove and use the facts that: – the angle subtended by an arc at the centre of a circle is twice the angle subtended at any point on the circumference – the angle in a semicircle is a right angle – angles in the same segment are equal – opposite angles of a cyclic quadrilateral sum to 180° – alternate segment theorem – the perpendicular from the centre of a circle to a chord bisect the chord
GM k	Use 2-D representations of 3-D shapes	• Use 2-D representations of 3-D shapes • Use isometric grids • Draw nets and show how they fold to make a 3-D solid • Understand and draw front and side elevations and plans of shapes made from simple solids • Given the front and side elevations and the plan of a solid, draw a sketch of the 3-D solid

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GM I	Describe and transform 2-D shapes using single or combined rotations, reflections, translations, or enlargements by a positive, fractional or negative scale factor and distinguish properties that are preserved under particular transformations	- Describe and transform 2-D shapes using single rotations - Understand that rotations are specified by a centre and an (anticlockwise) angle - Find the centre of rotation - Rotate a shape about the origin, or any other point - Describe and transform 2-D shapes using single reflections - Understand that reflections are specified by a mirror line - Identify the equation of a line of symmetry - Describe and transform 2-D shapes using single translations - Understand that translations are specified by a distance and direction (using a vector) - Translate a given shape by the vector $\begin{pmatrix} 2 \\ -3 \end{pmatrix}$ - Describe and transform 2-D shapes using enlargements by a positive and a negative or fractional scale factor - Understand that an enlargement is specified by a centre and a scale factor - Enlarge shapes using (0, 0) as the centre of enlargement - Enlarge shapes using centre other than (0, 0) - Find the centre of enlargement - Describe and transform 2-D shapes using combined rotations, reflections, translations, or enlargements - Distinguish properties that are preserved under particular transformations - Recognise that enlargements preserve angle but not length

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GM I	<i>(Continued)</i>	• Use congruence to show that translations, rotations and reflections preserve length and angle, so that any figure is congruent to its image under any of these transformations • Understand that distances and angles are preserved under rotations, reflections and translations so that any shape is congruent to its image • Describe a transformation
GM v	Use straight edge and a pair of compasses to carry out constructions	• Use straight edge and a pair of compasses to do standard constructions • Construct a triangle • Construct an equilateral triangle • Understand, from the experience of constructing them, that triangles satisfying SSS, SAS, ASA and RHS are unique, but SSA triangles are not • Construct the perpendicular bisector of a given line • Construct the perpendicular from a point to a line • Construct the perpendicular from a point on a line • Construct the bisector of a given angle • Construct angles of 60°, 90°, 30°, 45° • Draw parallel lines • Draw circles and arcs to a given radius • Construct a regular hexagon inside a circle • Construct diagrams of everyday 2-D situations involving rectangles, triangles, perpendicular and parallel lines • Draw and construct diagrams from given information

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GM w	Construct loci	- Construct: - a region bounded by a circle and an intersecting line - given distance from a point and a given distance from a line - equal distances from two points or two line segments - regions which may be defined by 'nearer to' or 'greater than' - Find and describe regions satisfying a combination of loci (NB: All loci restricted to two dimensions only)
GM x	Calculate perimeters and areas of shapes made from triangles, rectangles and other shapes	- Measure shapes to find perimeter or area - Find the perimeter of rectangles and triangles - Calculate perimeter and area of compound shapes made from triangles, rectangles and other shapes - Recall and use the formulae for the area of a triangle, rectangle and a parallelogram - Calculate areas of shapes made from triangles and rectangles - Calculate perimeters of compound shapes made from triangles and rectangles - Find the area of a trapezium - Find the area of a parallelogram - Find the surface area of simple shapes (prisms) using the formulae for triangles and rectangles, and other shapes
GM y	Calculate the area of a triangle using $\frac{1}{2}ab \sin C$	Calculate the area of a triangle given the length of two sides and the included angle

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GM z	Find circumferences and areas of circles	- Find circumferences of circles and areas enclosed by circles - Recall and use the formulae for the circumference of a circle and the area enclosed by a circle - Use $\pi \approx 3.142$ or use the π button on a calculator - Find the perimeters and areas of semicircles and quarter circles Calculate the lengths of arcs and the areas of sectors of circles Answers in terms of π may be required - Find the surface area of a cylinder
GM aa	Calculate volumes of right prisms and shapes made from cubes and cuboids	- Calculate volumes of right prisms, including the triangular prism, and shapes made from cubes and cuboids - Recall and use the formula for the volume of a cuboid - Find the volume of a cylinder - Use volume to solve problems
GM bb	Solve mensuration problems involving more complex shapes and solids	Solve problems involving more complex shapes and solids, including segments of circles and frustums of cones Find the surface area and volumes of compound solids constructed from cubes, cuboids, cones, pyramids, spheres, hemispheres, cylinders Examples:    Solve problems including examples of solids in everyday use Find the area of a segment of a circle given the radius and length of the chord

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GM cc	Use vectors to solve problems	• Understand and use vector notation • Calculate, and represent graphically, the sum of two vectors, the difference of two vectors and a scalar multiple of a vector • Calculate the resultant of two vectors • Solve geometrical problems in 2-D using vector methods • Apply vector methods for simple geometrical proofs